Graph-based Machine Learning

Hamid

Aalborg University Business School
Innovation, Knowledge, and Economic Development (IKE)
AI Denmark (AIDK)
hamidb@business.aau.dk
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Today's lecture overview

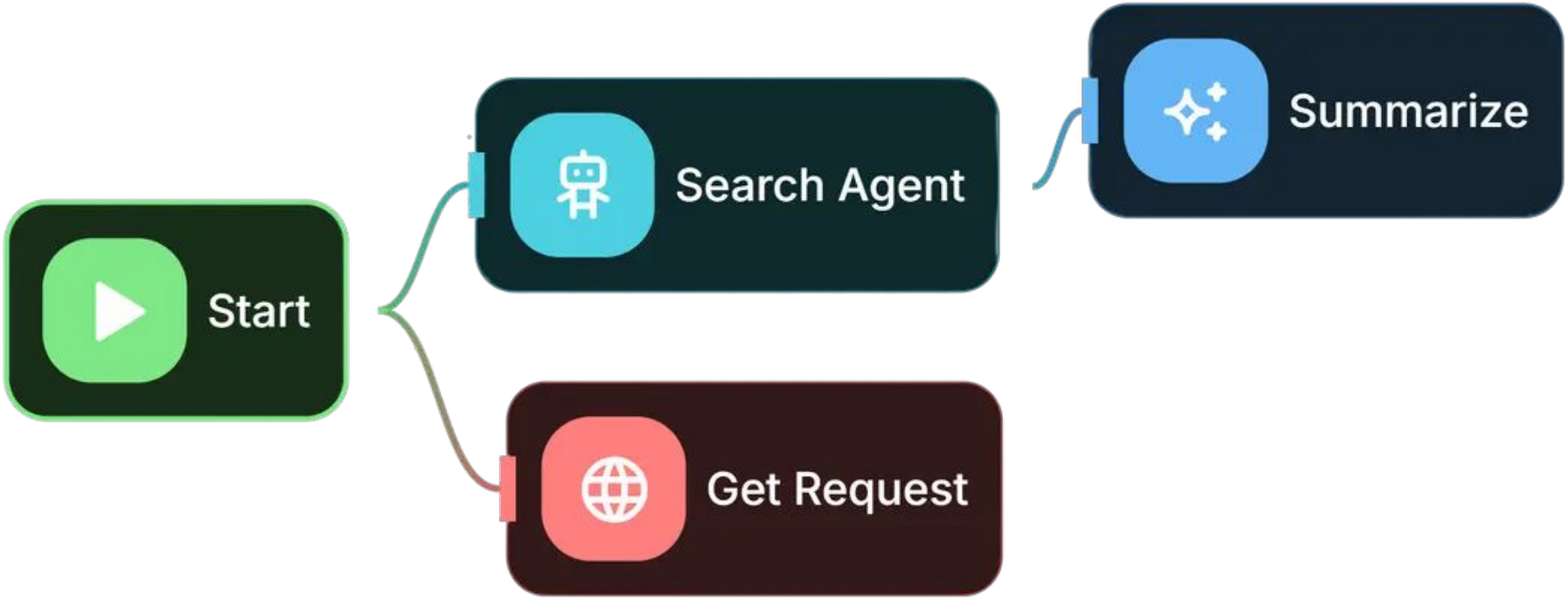
1. Recap
2. Prompt Tuning and QLoRA Fine tuning
3. Assignment 2 and 3
4. Knowledge Graph and GNN

Overview of the Upcoming Exam

1. **Presentation - Question & Answer (2 + 15 minutes)**
   This phase consists of two sections:

2. **Evaluation and Feedback (3 minutes)**
   The final 3 minutes will be dedicated to evaluation and feedback.

Flowise



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Stop fighting your tools

Langflow is a powerful tool to build and deploy AI agents and MCP servers. It comes with batteries included and supports all major LLMs, vector databases and a growing library of AI tools.

Get Started for Free

Star on GitHub

Input

Type something...

Model

llama-3.2

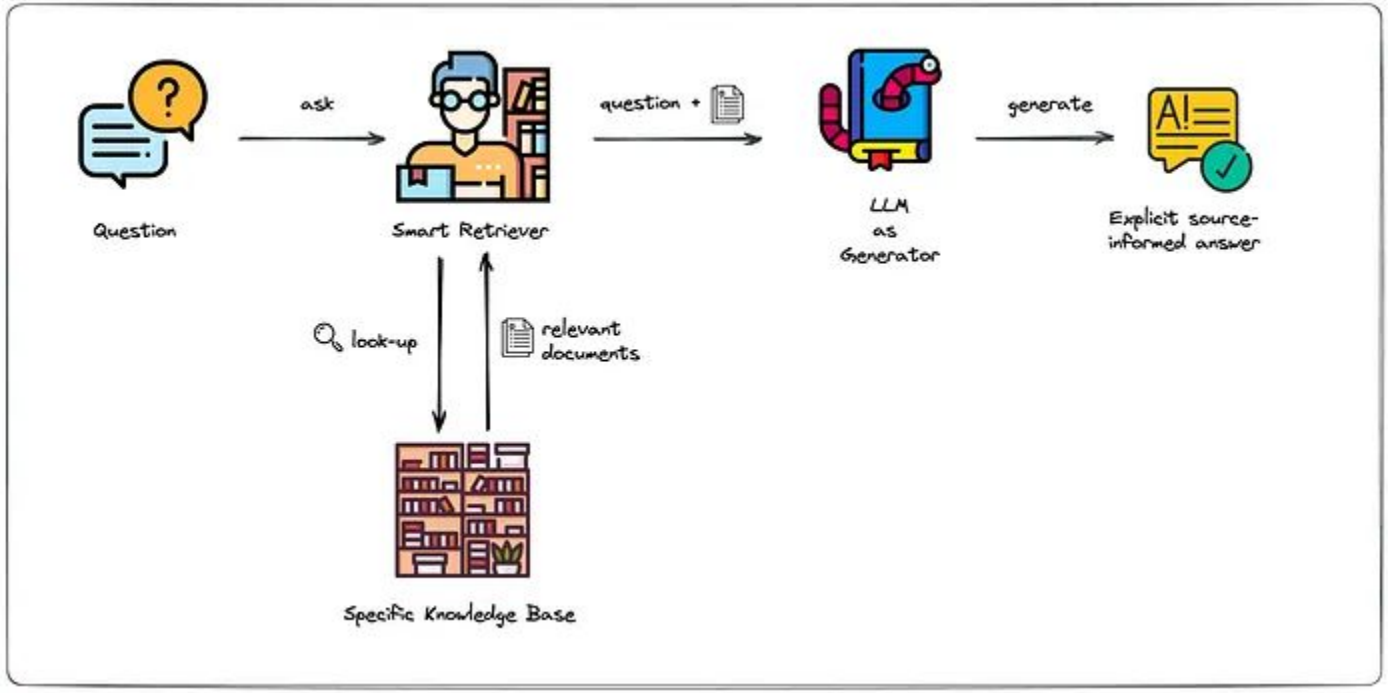
API Key

Type something...

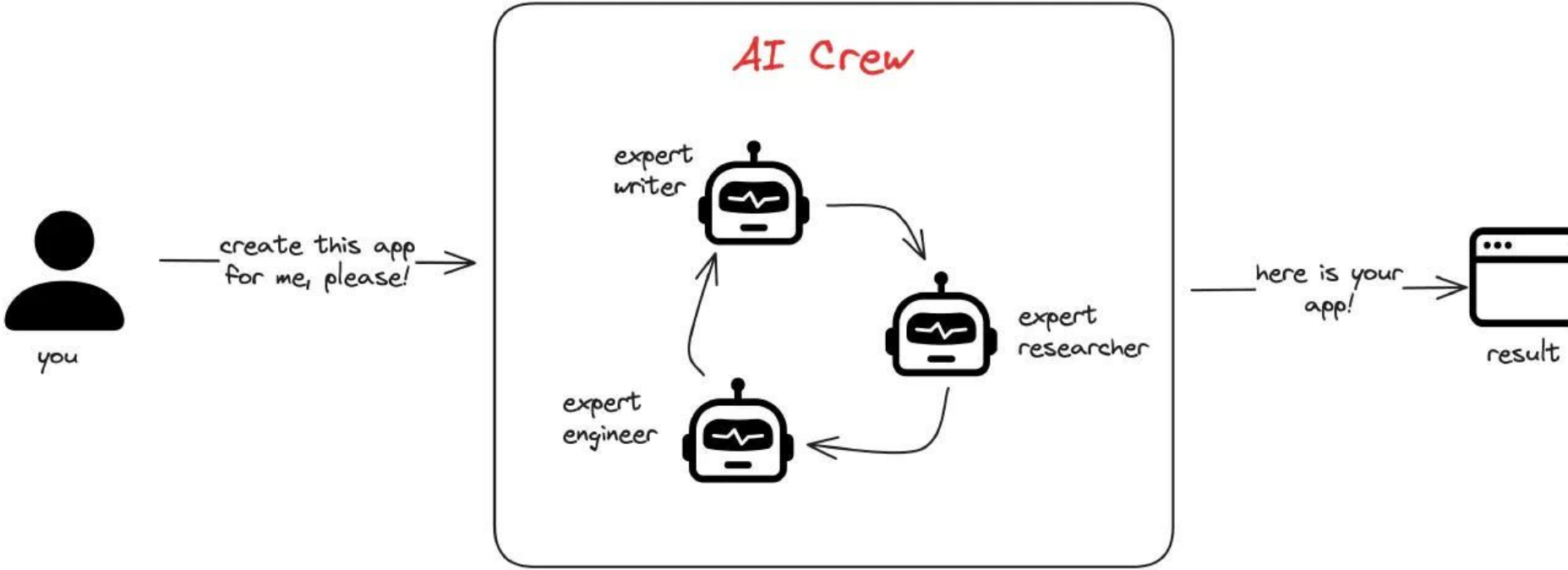
Temperature 0.5

Precise Creative

RAG + ChromaDB



CrewAI


AI Crew
expert writer
expert researcher
expert engineer
you
create this app for me, please!
here is your app!
result

Unlock AI's true impact across the SDLC. Explore key findings from Gartner®.

What is Agentic AI?

August 6, 2025

Discover how agentic AI helps software development teams increase productivity and focus on more strategic tasks.

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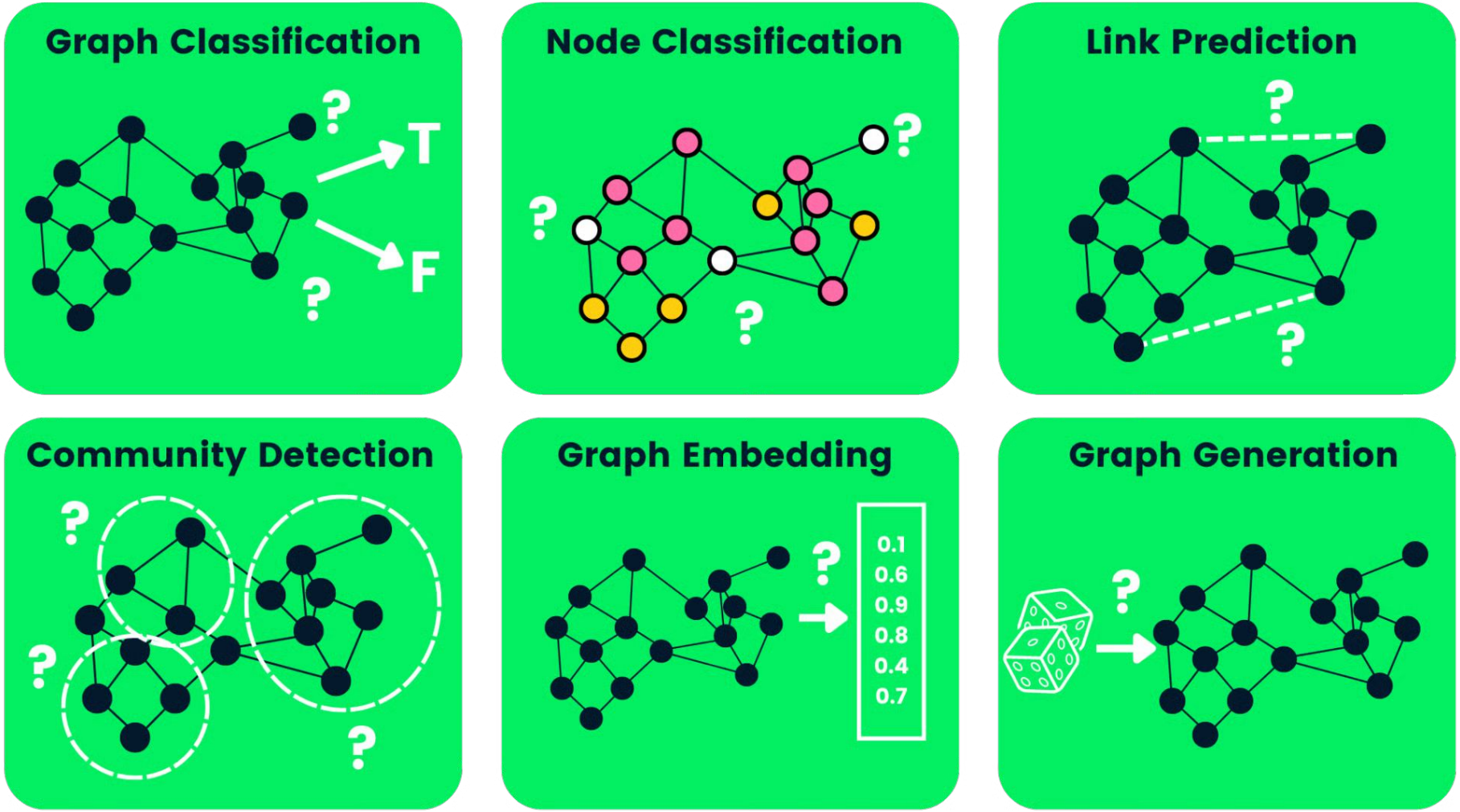
What is Agentic AI?

How does agentic AI work?

Agentic AI vs. generative AI

How to Validate LLMs?

	ways to grade answers			**the exam**
	Deterministic / Statistical	**LLM-as-a-Judge**	**Human-in-the-Loop**	**Benchmark Suites**
Full Name	NLP Metrics	Neural Evaluation / Automated Grading	Human Preference Evaluation	Standardized Benchmarking
Key Tools	BLEU, ROUGE, BERTScore, Exact Match (EM)	**DeepEval**, **Ragas**, **G-Eval**, Prometheus	**Chatbot Arena**, Argilla, Labelbox	**MMLU-Pro**, **GPQA**, **HumanEval**, **SWE-bench**
Data Needed	Ground Truth (Reference) answers	Rubric + LLM Judge + Context	Human Annotators + Comparison UI	Curated Test Sets (MCQs, Code, Math)
Core Philosophy	“Measure word overlap or semantic embedding similarity between output and a known answer.”	“Use a more capable model (Judge) to score a candidate model based on a custom rubric.”	“Humans are the final arbiters of helpfulness, safety, and nuance.”	“Standardized tests allow for head-to-head comparison across different providers.”


Graph Classification
?
T
F
?
Node Classification
?
?
?
Link Prediction
?
?
Community Detection
?
?
?
Graph Embedding
?
0.1
0.6
0.9
0.8
0.4
0.7
Graph Generation
?

Default Rubrics

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Reference-Free Rubrics (Default)

Score	Description
1	The response is entirely incorrect and fails to address any aspect of the user input.
2	The response contains partial accuracy but includes major errors or significant omissions.
3	The response is mostly accurate but lacks clarity, thoroughness, or minor details.
4	The response is accurate and clear, with only minor omissions or slight inaccuracies.
5	The response is completely accurate, clear, and thoroughly addresses the user input.

Reference-Based Rubrics

Score	Description
1	The response is entirely incorrect, irrelevant, or does not align with the reference.
2	The response partially matches the reference but contains major errors or omissions.
3	The response aligns with the reference overall but lacks sufficient detail or clarity.

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Domain-Specific Rubrics
Example
Custom Rubrics
With Retrieved Contexts
Convenience Classes
Default Rubrics
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Instance-Specific Rubrics
Example
With Reference and Contexts
Legacy API

LLM Evaluation - Metrics - ROUGE-L

Reference (human):

It is cold outside.

Generated output:

It is very cold outside.

LCS:
Longest common subsequence

ROUGE-L Recall: $= \frac{\text{LCS(Gen, Ref)}}{\text{unigrams in reference}} = \frac{2}{4} = 0.5$

ROUGE-L Precision: $= \frac{\text{LCS(Gen, Ref)}}{\text{unigrams in output}} = \frac{2}{5} = 0.4$

ROUGE-L F1: $= 2\,\frac{\text{precision x recall}}{\text{precision + recall}} = 2\,\frac{0.2}{0.9} = 0.44$

ROUGE

REF: *Today was sunny.*

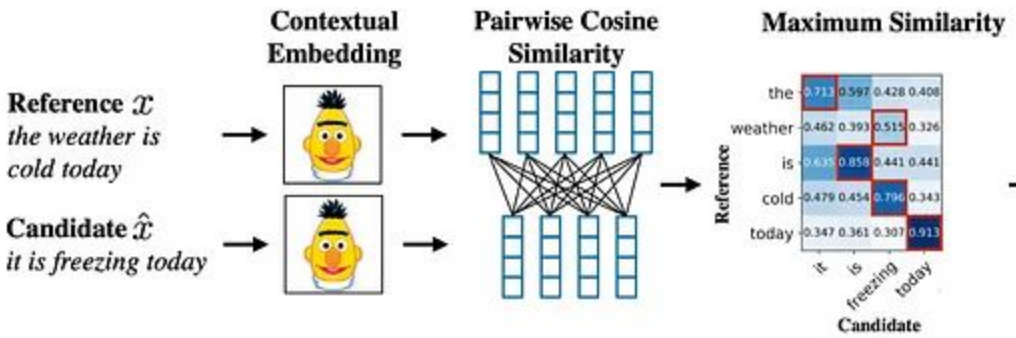
PRED: *Today the weather is sunny.*

$$\text{Precision} = \frac{\#\text{ overlapping words}}{\#\text{ candidate words}}$$

$$\text{Recall} = \frac{\#\text{ overlapping words}}{\#\text{ reference words}}$$

$$\text{ROUGE} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Introducing **BERTScore**



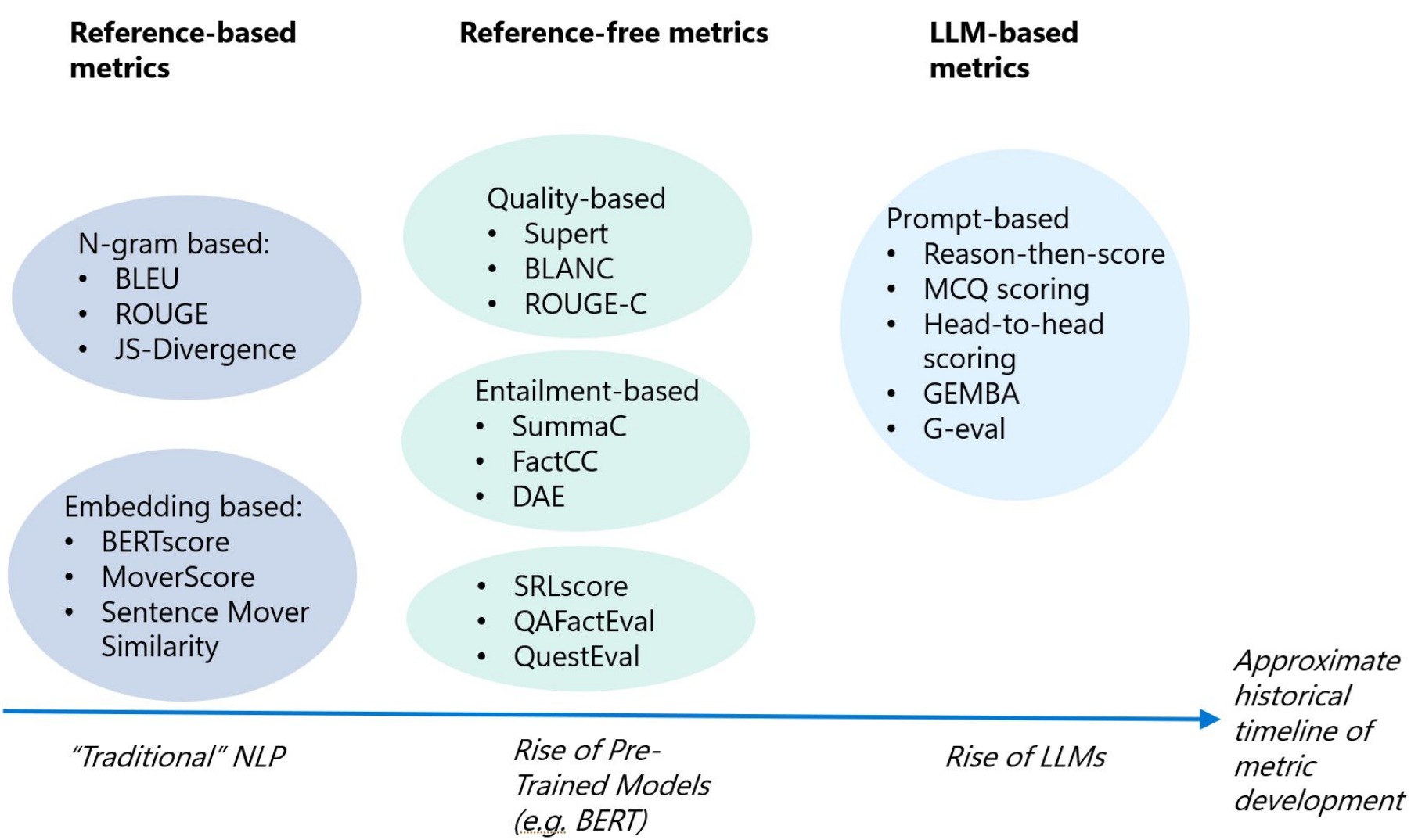
$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} \mathbf{x}_i^\top \hat{\mathbf{x}}_j$$

$$P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} \mathbf{x}_i^\top \hat{\mathbf{x}}_j$$

$$F_{\text{BERT}} = 2 \frac{P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}$$

Source: Bertscore: Evaluating text generation with bert

Code for Bertscore is available at https://github.com/Tiiiger/bert score


Reference-based metrics
Reference-free metrics
LLM-based metrics
N-gram based:
• BLEU
• ROUGE
• JS-Divergence
Embedding based:
• BERTscore
• MoverScore
• Sentence Mover Similarity
Quality-based
• Supert
• BLANC
• ROUGE-C
Entailment-based
• SummaC
• FactCC
• DAE
• SRLscore
• QAFactEval
• QuestEval
Prompt-based
• Reason-then-score
• MCQ scoring
• Head-to-head scoring
• GEMBA
• G-eval
"Traditional" NLP
Rise of Pre-Trained Models (e.g. BERT)
Rise of LLMs
Approximate historical timeline of metric development